Paper_ED_Cos_03 — CMB Acoustic Peaks as Substrate-Inherited Baryon-Photon Oscillator

Series: Wave-3 Generative Papers (Cosmology Arc — CMB acoustic-peak structure) **Status:** Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** within DCGT A→regime. Direct composition of Cos_01 (primordial spectrum) + Cos_02 (baryon density) + Cos_05 (late-time expansion) + Route A4 substrate-parameter inheritance + standard recombination + Boltzmann-code machinery via DCGT continuum bridge. **M3 via continuum-Friedmann inheritance** — Dyn_02 M2 RDE chain-class identification is *not* load-bearing (see Preamble item 6). **Date:** 2026-05-17 **Anchors:** Paper_087 (13 primitives); Paper_073 (DCGT); Paper_012 (substrate-c); Paper_027 (Newton's G); Paper_028 ($R_H = c/H_0$); Paper_ED_SC_4_9 (S1/S2/S3); Paper_ED_Cos_01 (M3 — Inflation); Paper_ED_Cos_02 (M3 — BBN); Paper_ED_Cos_05 (M3 — Dark Energy); Paper_ED_Dyn_02 (M2 — RDE, upstream-status only); Paper_095 (verdict grammar — §3.2.1 sub-labels). Substrate-research support memos cross-referenced in §2.

Label-class note: Audit-row sub-labels *substrate-parameter-INHERITED* and *IC-INHERITED* per Paper_095 §3.2.1.

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim closed-proof derivation of standard recombination physics (Saha equation; Peebles three-level model; Hyrec / Recfast code outputs) or standard CMB power-spectrum machinery (CAMB / CLASS Boltzmann codes) from substrate primitives. These continuum-side machineries are **INHERITED**; ED supplies the substrate-graph reading of $H(t)$ + cosmological-history-integration framework underlying them.
2. It does **not** sit within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). ED is in the substrate-ontology research program. Methodology per Paper_095 (form-FORCED / value-INHERITED).
3. It does **not** introduce new substrate primitives or paper-specific postulates. The CMB-acoustic content is composition of Cos_01 + Cos_02 + Cos_05 + Route A4 substrate-parameter-INHERITED scale + standard recombination + Thomson-scattering machinery + standard CMB power-spectrum codes via DCGT continuum bridge.
4. It does **not** claim numerical-precision derivation of specific peak positions, heights, or polarization observables (TT $\ell_1 \approx 220$, C_ℓ amplitudes, TE / EE cross-power spectra). Specific numerical values inherit from standard CAMB / CLASS Boltzmann-code outputs; ED supplies the substrate-graph mechanism producing the $H(t)$ + primordial-spectrum inputs.
5. It does **not** address CMB-spectral anomalies (low-ℓ deficit; hemispheric asymmetry; cold spot; etc.). These are standard-cosmology puzzles; ED inherits them at standard-cosmology level and does not supply substrate-graph mechanism specifically resolving them.
6. **Inheritance from Dyn_02 M2 status (architectural division).** Cos_03 inherits standard-cosmology continuum-Friedmann $H(t)$ via DCGT continuum bridge at M3-equivalent inheritance level. Dyn_02's substrate-graph RDE / MDE chain-class identifications (M2 with OPEN-HM-Q1 + OPEN-HM-Q2) remain OPEN at Dyn_02's level; **Cos_03 does not load-bear on these closures** because CMB-acoustic substrate-graph content (Route A4 H_0 + Θ_{ED} sensitivity channel for r_s, D_A, $\Omega_b h^2$) is independent of the chain-class identification for pre-recombination matter content. Same architectural-division

logic as Cos_02 + Dyn_01 + Dyn_04 + GW_01.

Abstract

The CMB acoustic-peak structure — first peak $\ell_1 \approx 220$, harmonic peaks $\ell_n \approx n\ell_1$, baryon-loading odd/even modulation, Silk-damping cutoff $\ell \gtrsim 1000$ — is identified at substrate-graph level as **the angular-projection at decoupling of the pre-recombination baryon-photon-fluid acoustic oscillator, governed by the substrate-graph-derived $H(t)$ and the Cos_01 primordial perturbation spectrum**. $H(t)$ inherits continuum-Friedmann via DCGT bridge (Cos_01 + Cos_02 + Cos_05 + Route A4 substrate-parameter inheritance); η_b via Cos_02; sound horizon r_s and angular-diameter distance D_A via standard-cosmology integration with substrate-parameter scaling; standard recombination + Thomson scattering + CAMB/CLASS Boltzmann codes inherited at textbook level. **Form-IDENTIFIED** via Cos_01 + Cos_02 + Cos_05 + DCGT composition. **Value-INHERITED**: r_s, D_A, $\Omega_b h^2$, H_0 via Route A4; specific peak positions/heights via CAMB / CLASS. **IC-INHERITED**: primordial spectrum (Cos_01), recombination physics, $\Omega_b h^2$ (Cos_02 + CMB cross-consistency). Substrate-c preserved. **Verdict per Paper_095: M3** (form-IDENTIFIED + value-INHERITED + IC-INHERITED); Dyn_02 M2 *not* load-bearing (Preamble item 6). **F1 (sharpest):** CMB peak positions or peak-height ratios significantly inconsistent with substrate-graph r_s/D_A under Route A4 substrate-parameter scaling — beyond Planck ~0.1%-precision on ℓ_1 — refutes the composition. **Empirical anchors are exceptionally strong:** Planck 2018 TT/TE/EE spectra match ΛCDM at <1% across all measured peaks. **Distinguishing ED contribution:** Route A4 substrate-parameter $\Theta_{\rm ED}$ scaling channel for r_s, D_A, $\Omega_b h^2$ supplies ED's structural link to the Hubble-tension observational target; whether the tension resolves via substrate-parameter tightening (RA-OPEN-4a) or standard intermediate-redshift physics modification is OPEN and not load-bearing for Cos_03's M3 verdict. Enables Paper_ED_Cos_04 (linear structure formation); strengthens CMB B-mode polarization targets (BICEP / Simons Observatory / CMB-S4 / LiteBIRD) via Cos_01 inheritance.

---

§1 Statement of Result

The CMB acoustic-peak structure is the substrate-graph **angular projection at decoupling of the pre-recombination baryon-photon-fluid acoustic oscillator**, with peak positions

$$\ell_n \approx n\pi \frac{D_A(z_{\rm rec})}{r_s(z_{\rm rec})}$$

where $r_s = \int_0^{t_{\rm rec}} c_s(t)/a(t)\,dt$ is the comoving sound horizon at decoupling, $c_s = 1/\sqrt{3(1+R)}$ is the baryon-photon-fluid sound speed with $R = 3\rho_b/(4\rho_\gamma)$, and $D_A(z_{\rm rec})$ is the angular-diameter distance to the decoupling surface.

Three temporal regimes (standard CMB-acoustic phenomenology, substrate-side reading):

Regime	Era	Substrate-graph content
Pre-recombination	$z \gtrsim 10^3$	Baryon-photon fluid as ED-substrate acoustic oscillator on FRW background; sound horizon $r_s(t)$ grows; $H(t)$ at radiation-dominated $\to$ matter-radiation-equality crossing
Recombination decoupling	$z_{\text{rec}} \approx 1090$	Photons free-stream at $T \approx 0.26$ eV; acoustic snapshot frozen at last-scattering surface
Post-recombination	$z < 1090$	Free-streaming photons + late-time gravitational potentials (ISW); angular projection at $D_A(z_{\text{rec}})$ supplies observed C_ℓ

FORM: acoustic peaks = baryon-photon-fluid oscillator with sound horizon / angular-diameter-distance ratio — **IDENTIFIED** via Cos_01 + Cos_02 + Cos_05 + Route A4 + standard recombination + Boltzmann-code composition. Zero pure-D rows.

VALUE: r_s + D_A + $\Omega_b h^2$ + H_0 all inherit Route A4 substrate-parameter-INHERITED scaling at Paper_027-analog tier; specific peak positions + heights inherit standard CAMB / CLASS Boltzmann-code outputs.

IC: primordial perturbation spectrum (nearly scale-invariant; $n_s \approx 0.965$) inherited from Cos_01 inflation residual; baryon density $\Omega_b h^2 \approx 0.0224$ inherited from Cos_02 BBN + CMB cross-consistency; recombination physics inherited from standard cosmology.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime per §4 audit table.

---

§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02, P03, P09, P10, P11, P12, P13.

Upstream papers: - Paper_012 (P-RB-1 substrate-c bound) - Paper_027 (Newton's G — substrate-parameter inheritance) - Paper_028 (cosmic decoupling surface $R_H = c/H_0$) - Paper_073 (DCGT hydrodynamic-window coarse-graining — supplies continuum-Friedmann inheritance) - Paper_ED_SC_4_9 (saddle classification) - Paper_ED_Cos_01 (M3 — supplies inflation-residual primordial perturbation spectrum + nearly-scale-invariant tensor / scalar power) - Paper_ED_Cos_02 (M3 — supplies BBN-derived baryon density $\Omega_b h^2$ + η_b entering baryon-loading R) - Paper_ED_Cos_05 (M3 — supplies late-time expansion history + Route A4 H_0 for $D_A(z_{\text{rec}})$ inheritance) - Paper_ED_Dyn_02 (M2 — RDE chain-class identification at upstream status; non-load-bearing for Cos_03 via continuum-Friedmann inheritance) - Update_ED_SC_4x_Arc_M3 (six-projection arc-wide M3 upgrade post-Route-A) - Standard recombination machinery (Saha equation; Peebles three-level model; Hyrec / Recfast codes) — textbook inheritance - Standard Thomson-scattering + baryon-photon-fluid machinery — textbook inheritance - Standard CMB Boltzmann codes (CAMB / CLASS) — textbook inheritance for specific peak positions, heights, polarization observables

Substrate-research support: - Memo_ED_RouteA_A4_Audit + Memo_ED_RouteA_MultiRouteConvergen (Route A4 substrate-parameter-INHERITED closure) - Memo_ED_Q1Q2_JointClosure_Construct + Memo_ED_ChainClass_C3_Construct + Audit

No new substrate primitives. No paper-specific postulates.

---

§3 Derivation

§3.1 Baryon-photon fluid as ED-substrate acoustic oscillator

Pre-recombination, baryons (electrons + nuclei from Cos_02 BBN) are tightly coupled to photons via Thomson scattering; the combined baryon-photon fluid behaves as a single relativistic fluid with sound speed $c_s(t) = 1/\sqrt{3(1+R(t))}$ where $R(t) = 3\rho_b(t)/(4\rho_\gamma(t)) \propto a(t)$.

Substrate-graph composition: standard baryon-photon coupling + substrate-graph-inherited $H(t)$ + Cos_02 baryon density $\Omega_b h^2$ via η_b. The substrate-graph identification of baryon-photon fluid at substrate level is standard relativistic-fluid territory (mixed-signature substrate state at pre-recombination temperatures); Paper_ED_Dyn_02 supplies the substrate-graph RDE identification at M2 upstream status (non-load-bearing for Cos_03's continuum-side inheritance).

Acoustic perturbations propagate at c_s from the end of inflation (Cos_01 supplies the inflation-residual primordial spectrum as initial conditions for the acoustic oscillator) until recombination decoupling.

§3.2 Sound horizon r_s from ED $H(t)$

The comoving sound horizon at decoupling:

$$r_s(z_{\rm rec}) = \int_0^{t_{\rm rec}} \frac{c_s(t)}{a(t)} dt = \int_{z_{\rm rec}}^{\infty} \frac{c_s(z)\, dz}{H(z)(1+z)}$$

where $H(z)$ inherits Cos_01 + Cos_02 + Cos_05 expansion-history framework + Route A4 substrate-parameter-INHERITED scale. Standard cosmological evolution from inflation through RDE through matter-radiation equality ($z_{\rm eq} \approx 3400$) to recombination ($z_{\rm rec} \approx 1090$) supplies the $H(z)$ profile.

Numerical result (standard cosmology + Route A4 H_0): $r_s(z_{\rm rec}) \approx 147$ Mpc comoving. Substrate-parameter-INHERITED scaling: $r_s \propto 1/H_0 \propto \Theta_{\rm ED}^{-1/2}$ via Route A4 chain.

§3.3 Peak positions $\ell_n \approx n\pi D_A/r_s$

Angular-diameter distance to the recombination surface:

$$D_A(z_{\rm rec}) = \frac{c}{(1+z_{\rm rec})} \int_0^{z_{\rm rec}} \frac{dz}{H(z)}$$

inherits Cos_05 + Cos_01 + Route A4 expansion-history framework. Numerical result: $D_A(z_{\rm rec}) \approx$ 13 Mpc proper (after the $1/(1+z_{\rm rec})$ factor — corresponding to comoving distance $\approx$ 14 Gpc).

Acoustic peak positions in the CMB angular power spectrum:

$$\ell_n \approx n\pi \frac{D_A(z_{\rm rec})}{r_s(z_{\rm rec})}$$

For $n = 1$: the simple analytic formula gives $\ell_1 \approx \pi \cdot 14000/147 \approx 300$; the observed $\ell_1 \approx 220$ requires standard cosmological corrections (driving effects, projection effects, exact recombination physics) supplied by CAMB / CLASS Boltzmann-code calculations. **ED's full prediction equals the standard ΛCDM CAMB output**, not the simple analytic estimate — the substrate-graph framework inherits the full standard CMB-physics calculation; the simple-formula number above is for illustration of the dominant geometric scaling only, not ED's prediction.

Key cancellation: the ratio D_A/r_s is partially insensitive to H_0 (both scale as $1/H_0$ at leading order); the residual sensitivity to H_0 is a known feature of CMB acoustic-peak observables, providing the H_0 constraint from CMB observations.

§3.4 Baryon loading and odd/even peak modulation

The baryon-loading parameter $R = 3\rho_b/(4\rho_\gamma)$ at recombination ($R_{\rm rec} \approx 0.6$ for $\Omega_b h^2 \approx 0.0224$) controls the asymmetric oscillation amplitude: in the rest frame of the baryon-photon fluid, the gravitational potential acts asymmetrically, suppressing compression-phase oscillations (which produce odd-numbered peaks $\ell_1, \ell_3, \ldots$) less than rarefaction-phase oscillations (even-numbered peaks $\ell_2, \ell_4, \ldots$). Standard prediction: odd peaks enhanced relative to even peaks by factor $\sim (1+6R)$.

Substrate-graph composition: $\Omega_b h^2$ inherits Cos_02 BBN + CMB cross-consistency; baryon-loading inherits standard acoustic-physics calculation; substrate-graph framework supplies the $H(t)$ + primordial-spectrum + baryon-density inputs to the standard calculation.

Observed: Planck 2018 measures the second peak relative to first, third relative to second, fourth relative to third, etc., to ~1% precision — consistent with standard prediction at $\Omega_b h^2 \approx 0.0224$.

§3.5 Silk damping (diffusion scale)

Photon diffusion during the recombination transition smooths fluctuations on scales below the Silk damping scale:

$$k_D \sim \frac{1}{\sqrt{\int dt/[n_e \sigma_T a^2]}}$$

where n_e is the free-electron density (set by recombination physics) and σ_T is the Thomson cross-section. The corresponding angular scale: $\ell_D \sim k_D D_A(z_{\rm rec}) \sim 1500$.

For $\ell \gtrsim \ell_D$, C_ℓ is exponentially suppressed (Silk damping tail). Substrate-graph composition: standard diffusion-physics calculation + substrate-graph-inherited $H(t)$ + Cos_02 baryon density.

§3.6 Sensitivity to Θ_{ED} or τ_{V_5}

Substrate-parameter Θ_{ED} enters CMB acoustic observables through Route A4's identification $H_0 \propto \Theta_{\mathrm{ED}}^{1/2}$:

- $r_s \propto 1/H_0 \propto \Theta_{\mathrm{ED}}^{-1/2}$
- $D_A \propto 1/H_0 \propto \Theta_{\mathrm{ED}}^{-1/2}$
- $\ell_n \propto D_A/r_s$: dominant scaling cancels; **residual sensitivity** $\propto \Omega_m^{0.5}$ or similar weak combination (standard CMB-cosmology analysis)
- $\ell_D \propto k_D \cdot D_A$: k_D has H_0-dependence through recombination physics; $D_A \propto 1/H_0$; net ℓ_D has some H_0 sensitivity

Hubble tension framing (honest scope): CMB-derived $H_0 \approx 67.4$ km/s/Mpc (Planck 2018) vs SH0ES local $H_0 \approx 73$ km/s/Mpc (~5 discrepancy as of 2024). Standard ΛCDM also predicts a single H_0 at the present epoch under a given parameter set — the tension is between two measurement channels of the same parameter, not between two parameter values, and ED's Route A4 channel does not naturally resolve it. **Route A4 supplies ED's structural link to the Hubble-tension observational target via the substrate-parameter scaling channel** (the same Θ_{ED} enters r_s, D_A, and H_0); whether the tension resolves via substrate-parameter tightening (RA-OPEN-4a closure direction) or standard intermediate-redshift physics modification (early dark energy, new recombination physics — standard-cosmology territory, not substrate-graph-specific) is OPEN and **not load-bearing for Cos_03's M3 verdict**. If the tension is genuine and persists, either pathway requires substrate-research-frontier or standard-cosmology-level work; neither aggravates the Cos_03 chain.

Substrate-parameter values $\Theta_{\mathrm{ED}}^{\mathrm{Planck-units}} \approx 10^{-122}$ at primitive level; tightening via RA-OPEN-4a/4b/4c-explicit substrate-research-frontier, not blocking for Cos_03's M3 verdict.

§3.7 Comparison to ΛCDM CMB acoustic physics

Element	ΛCDM standard	ED substrate-graph reading	Status
Baryon-photon fluid coupling + Thomson scattering	Standard QED + relativistic fluid dynamics	Same; substrate-graph baryon-photon coupling inherits	**Inherited**
Sound horizon r_s	Standard Friedmann integration	Substrate-graph $H(t)$ via Cos_01 + Cos_02 + Cos_05 + Route A4 inheritance	**D-via-I via continuum-Friedmann inheritance**
Angular-diameter distance $D_A(z_{\mathrm{rec}})$	Standard Friedmann integration	Same; substrate-graph $H(z)$ inheritance	**D-via-I**
Peak positions $\ell_n \approx n\pi D_A/r_s$	Standard CMB physics	Same; substrate-graph r_s, D_A inheritance	**D-via-I**
Baryon-loading + odd/even modulation	Standard acoustic physics; $\Omega_b h^2$ from BBN+CMB	Same; $\Omega_b h^2$ inherits Cos_02 BBN + CMB cross-consistency	**D-via-I**

Element	ΛCDM standard	ED substrate-graph reading	Status
Silk damping scale ℓ_D	Standard photon-diffusion physics	Same; substrate-graph-inherited $H(t)$ + Cos_02 baryon density	**D-via-I**
Substrate-parameter Θ_{ED} sensitivity	None	$H_0 \propto \Theta_{\text{ED}}^{1/2}$ via Route A4 → CMB observables inherit substrate-parameter scaling	**Distinguishing content**
Hubble tension	Observational/methodological puzzle	Same; substrate-research-frontier tightening direction	**Inherited puzzle** + distinguishing content via Route A4 channel

---

§4 Audit Table

#	Step	Label	Notes
1	P02, P03, P09, P10, P11, P12, P13	**P**	Paper_087.
2	Upstream inheritance block	**I**	Paper_012; Paper_027; Paper_028; Paper_073; Paper_ED_SC_4_9; Paper_ED_Cos_01 (M3); Paper_ED_Cos_02 (M3); Paper_ED_Cos_05 (M3); **Paper_ED_Dyn_02 (M2 — upstream, non-load-bearing)**; Update_ED_SC_4x_Arc_M3.
3	Standard recombination machinery (Saha; Peebles; Hyrec / Recfast)	**I**	Textbook inheritance via DCGT continuum bridge.

#	Step	Label	Notes
4	Standard Thomson-scattering + baryon-photon-fluid machinery	**I**	Textbook inheritance.
5	Standard CMB Boltzmann codes (CAMB / CLASS)	**I**	Textbook inheritance for specific peak positions, heights, polarization.
6	Baryon-photon fluid as ED-substrate acoustic oscillator	**D-via-I**	§3.1. Composition of standard fluid coupling + substrate-graph $H(t)$ + Cos_02 $\Omega_b h^2$.
7	Sound horizon $r_s(z_{\mathrm{rec}}) = \int (c_s/a)dt$ inherits substrate-graph $H(t)$	**D-via-I**	§3.2. Composition of standard cosmology integration + Cos_01 + Cos_02 + Cos_05 + Route A4.
8	Angular-diameter distance $D_A(z_{\mathrm{rec}})$ inherits substrate-graph $H(z)$	**D-via-I**	§3.3. Same composition.
9	Peak positions $\ell_n \approx n\pi D_A/r_s$	**D-via-I**	§3.3. Geometric construction + rows 7, 8.
10	Baryon-loading $R = 3\rho_b/(4\rho_\gamma)$ + odd/even peak modulation	**D-via-I**	§3.4. Composition of standard acoustic physics + Cos_02 η_b.
11	Silk damping scale $\ell_D \sim 1500$ from photon-diffusion physics	**D-via-I**	§3.5. Standard diffusion + substrate-graph $H(t)$ + Cos_02.
12	Specific peak positions, heights, TE/EE cross-spectra (CAMB / CLASS outputs)	**I**	§3.3 + §3.4. Standard Boltzmann-code inheritance.
13	Substrate-parameter Θ_{ED} sensitivity: $r_s, D_A \propto \Theta_{\mathrm{ED}}^{-1/2}$; ℓ_n has residual weak sensitivity through cancellation	**D-via-I**	§3.6. Composition of Route A4 + standard CMB-cosmology analysis.
14	Substrate-c bound preserved throughout	**D-via-I**	§3.1. Paper_012 + Paper_089.

#	Step	Label	Notes
15	H_0 at Route A4 substrate-parameter-INHERITED level	**I**	Sub-type per Paper_095 §3.2.1: *substrate-parameter-INHERITED*. §3.2, §3.3, §3.6. Paper_027-analog tier per Memo_RouteA_MultiRouteConverge option (ii).
16	Primordial perturbation spectrum ($n_s \approx 0.965$, scalar / tensor power) inherited from Cos_01 inflation residual	**I**	Sub-type per Paper_095 §3.2.1: *IC-INHERITED*. §2.IC. Cos_01 inflation framework + standard inflationary-perturbation theory.
17	Recombination physics ($z_{\text{rec}} \approx 1090$, $T_{\text{rec}} \approx 0.26$ eV) inherited from standard cosmology	**I**	Sub-type per Paper_095 §3.2.1: *IC-INHERITED*. §2.IC. Standard recombination codes.
18	$\Omega_b h^2$ inherited from Cos_02 BBN + CMB cross-consistency	**I**	Sub-type per Paper_095 §3.2.1: *IC-INHERITED*. §2.IC. Cos_02 + Planck η_b inheritance.
19	Hubble tension (CMB vs SH0ES H_0): substrate-research-frontier tightening direction	**OPEN** (non-load-bearing)	§3.6. Inherited observational puzzle; substrate-research-frontier direction.
20	Verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime	**A→position**	Per Paper_095 §3.3 line 77. Five-anchor verdict-sync verified.

Audit summary: zero pure-D rows; **nine substantive D-via-I rows** (6–11, 13, 14); **five I rows** (2, 3, 4, 5, 12); one substrate-parameter-INHERITED row (15); three IC-INHERITED rows (16, 17, 18); one non-load-bearing OPEN (19 — Hubble tension); one P row (1); one A→position verdict row (20). **No load-bearing OPEN items.** Upstream Dyn_02 M2 (RDE chain-class identification) inherited at upstream status per Preamble item 6 without load-bearing. No new substrate primitives; no paper-specific postulates.

§5 Falsification Criteria

- **F1 (sharpest):** Observed CMB peak positions ℓ_n or peak-height ratios significantly inconsistent with substrate-graph-derived r_s/D_A ratio under Route A4 H_0 substrate-parameter scaling — beyond Planck-level observational precision (~0.1% on ℓ_1). Refutes §3.2-§3.4 + audit rows 7-10. Currently consistent: Planck 2018 TT/TE/EE spectra match ΛCDM at all measured precision.
- **F2:** Observed odd/even peak-ratio modulation significantly inconsistent with ED-inherited baryon loading via Cos_02 + standard acoustic physics. Refutes §3.4 + audit row 10.
- **F3:** Observed Silk damping scale ℓ_D significantly inconsistent with substrate-graph-derived $H(t)$ + standard diffusion physics. Refutes §3.5 + audit row 11.
- **F4 (substrate-research-frontier-conditional):** Substrate-graph proof that the Route A4 H_0 substrate-parameter scaling channel produces structurally inconsistent r_s, D_A, $\Omega_b h^2$ predictions when simultaneously constrained by Planck TT/TE/EE + BBN η_b + SH0ES H_0 at observed precision. **Empirical refutation not currently available** (depends on RA-OPEN-4a closure + Hubble-tension observational resolution). Refutes §3.6 + audit rows 13, 15.

---

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

CMB acoustic peaks in ED are the substrate-graph reading of standard pre-recombination baryon-photon acoustic oscillation + angular projection at decoupling: peak positions $\ell_n \approx n\pi D_A/r_s$ with sound horizon r_s + angular-diameter distance D_A inheriting substrate-graph $H(t)$ via continuum-Friedmann inheritance from Cos_01 + Cos_02 + Cos_05 + Route A4 substrate-parameter-INHERITED scaling. Baryon-loading R inherits Cos_02 $\Omega_b h^2$; Silk damping inherits standard diffusion physics; specific peak observables inherit standard CAMB / CLASS Boltzmann-code outputs. Substrate-c bound preserved throughout pre-recombination evolution + post-recombination free-streaming.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window per Paper_095. Nine D-via-I rows; five I rows; one substrate-parameter-INHERITED row (Route A4); three IC-INHERITED rows (primordial spectrum + recombination physics + baryon density); zero pure-D rows; no paper-specific postulates; no new substrate primitives; **no load-bearing OPEN items**. One non-load-bearing OPEN (Hubble tension; inherited observational puzzle). Upstream Dyn_02 M2 (RDE chain-class identification pending Route C1) inherited at upstream status per Preamble item 6; not load-bearing for Cos_03's net M3 via continuum-Friedmann inheritance. **Empirical-anchor strength:** Planck 2018 TT/TE/EE matches ΛCDM at <1% precision across all measured peaks — one of the most precise empirical anchors in physics. The Route A4 substrate-parameter Θ_{ED} scaling channel for r_s, D_A, $\Omega_b h^2$ is the genuine ED-distinguishing content; future precision H_0 measurements (LIGO standard-sirens; JWST distance-ladder refinement; CMB-S4 / LiteBIRD precision) provide ED-vs-standard discriminator pathways via the substrate-parameter-INHERITED level.

Cross-arc consequences. Upstream, Cos_03 inherits unchanged from Cos_01 (M3), Cos_02 (M3), Cos_05 (M3), Dyn_02 (M2 — chain-class OPENs not aggravated per Preamble item 6). Downstream, Cos_03 *enables* Paper_ED_Cos_04 (linear structure formation) via composition with Cos_05 + standard cosmology (matter-radiation transfer function inherits Cos_03's CMB-acoustic framework); *strengthens* the SCBU cosmic-decoupling projection via the observed $\ell_1 \approx 220$ empirical anchor at substrate-parameter-INHERITED level; *strengthens* the empirical-prediction sector through the Route A4 Hubble-tension structural-link channel (whether the tension resolves via substrate-parameter tightening or standard intermediate-redshift physics is OPEN, not load-bearing); *strengthens* the CMB B-mode polarization observational targets (BICEP, Simons Observatory, CMB-S4, LiteBIRD) via Cos_01 inflation-residual primordial-tensor inheritance composed through Cos_03. Whether each downstream paper closes at M3 depends on its own substrate-graph chain; not forecast here.

End Paper_ED_Cos_03.